Coupled Oscillators

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Coupled oscillators are more complex than a single oscillator. In this experiment, we studied linear coupled oscillators, where the forces between the coupled objects are proportional to the displacement of the equilibrium position. We studied the following 3 modes and their angular frequency, amplitude and used eigenvalue method find the allowed frequencies: symmetric mode, antisymmetric mode, and a mixed mode.

1. INTRODUCTION

Coupled oscillators are fundamental to understanding dynamical systems, from molecular vibrations to electronic circuits. This experiment explores the physics behind the linear coupled oscillators—a system where two masses interact through springs, exhibiting vibrational modes governed by symmetry and energy exchange. While previous studies have established theoretical frameworks for normal modes, experimental validation of these theory under non-ideal scenarios, such as varying spring constants or unequal masses are still meaningful for bridging theory and real-world applications.

In this experiment, we investigate how two masses connected by springs oscillate in phase or out of phase, governed by Newtonian mechanics and Hooke's law. The system exhibits two distinct special vibrational frequencies: a symmetric mode, where masses move synchronously, and an antisymmetric mode, where they oscillate in opposite direction. These behaviors depend on spring constants and mass, modeled by coupled differential equations. However, real-world complexities—such as nonlinear spring constant impacts the analysis of the theory analysis.

The main goal of this experiment is to determine the angular frequencies of coupled oscillators and compare them to theoretical predictions derived from eigenvalue analysis. Using a PASCO air-track system with optical encoders and LabVIEW data acquisition, we measure position-time profiles and perform Fourier analyses on interface to identify dominant frequencies. Additionally, we explore how unequal masses or non-uniform spring constants affect system dynamics.

2. THEORY [1]

The following system consists of 2 masses and 3 springs and they are attached to over a frictionless horizontal track, with 2 immovable ends. x_1 is displacement of m_1 from the equilibrium, x_2 is displacement of m_2 from the equilibrium.

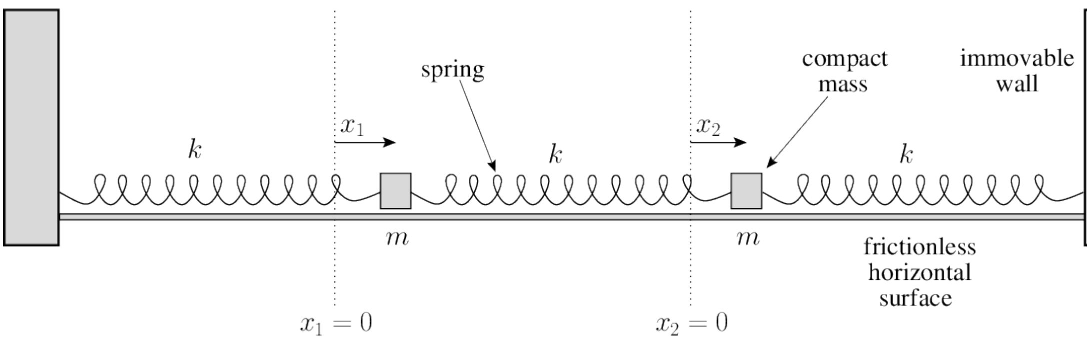

FIG. 1: Two-degree-of-freedom mass-spring system [1]

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The equations of motion of the two masses are thus

$$m_1 \frac{d^2 x_1}{dt^2} = -k_1 x_1 + k_2 (x_2 - x_1) \tag{1}$$

$$m_2 \frac{d^2 x_2}{dt^2} = -k_3 x_2 - k_2 (x_2 - x_1) \tag{2}$$

We are interested in the position of the masses as a function of time. As we are dealing with springs and oscillations, we can assume that the solution to this system will be in the general form of

$$x_i = B_i e^{i\omega t + \phi_i} \tag{3}$$

, where B is the amplitude of the motion, ϕ is the initial phase, and the allowed frequencies ω remain to be determined

$$-m_1 \omega^2 x_1 = -k_1 x_1 + k_2 (x_2 - x_1) \leftrightarrow m_1 \omega^2 x_1 - k_1 x_1 + k_2 (x_2 - x_1) = 0 \tag{4}$$

$$-m_1 \omega^2 x_1 = -k_1 x_1 + k_2 (x_2 - x_1) \leftrightarrow m_1 \omega^2 x_1 - k_1 x_1 + k_2 (x_2 - x_1) = 0 \tag{5}$$

One of our goals is to find the allowed frequencies of the system. In other words, our goal is to find the characteristics of a matrix *such that* the system of equations is satisfied (i.e. the motion equations). Mathematically, the system of equations should have non-trivial solutions, where determinant should be zero. By reorganizing Eq.4 and 5, we get a matrix of the system of equations as following:

$$\begin{pmatrix} m_1\omega^2 - k_1 - k_2 & k_2 \\ k_2 & m_2\omega^2 - k_2 - k_3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0 \tag{6}$$

$$\omega_1 = \sqrt{\frac{3k}{m}} \tag{7}$$

$$\omega_2 = \sqrt{\frac{k}{m}} \tag{8}$$

for the allowed frequencies and the physical solution to the positions:

$$x_1(t) = B_1 \cos(\omega_1 t + \phi_1) + B_2 \cos(\omega_2 t + \phi_2) \tag{9}$$

$$x_2(t) = -B_1 \cos(\omega_1 t + \phi_1) + B_2 \cos(\omega_2 t + \phi_2) \tag{10}$$

where B_i and ϕ_i can be determined using the initial conditions (t = 0). Equations 9 and 10 show that there are three different kinds of motion for this system.

- Anti-symmetric. Both masses oscillate at ω_1, meaning that $B_1 \neq 0$ and $B_2 = 0$. In this case, the masses move opposite each other.
- Symmetric mode. Both masses can be oscillating at ω_2, meaning that $B_1 = 0$ and $B_2 \neq 0$. In this case the masses move together.
- Mixed mode. The more general case is when both $B_1 \neq 0$ and $B_2 \neq 0$. Both frequencies will be present in the motion.

3. EXPERIMENTAL METHODS

3.1. Experimental Methods

The experimental setup consists of a one-dimensional coupled oscillator system on a PASCO air-track, designed to minimize friction. A schematic of the apparatus is shown in Fig. 2, where Cart 1 (mass m_1) and Cart 2 (mass m_2) are coupled via three springs: k_1 and k_3 anchor the carts to fixed boundaries, and k_2 connects the two carts. Position and Amplitude data is acquired using a Mylar encoder strip attached to Cart 1, measured and monitored by optical decoders (HEDS 9200-R00) interfaced with LabVIEW program through PCI-1200 digital I/O.

Before measuring the system, the spring constants k_1, k_2, and k_3 were determined using Hooke's law:

- $k_1 = 3.60441\, N/m$,
- $k_2 = 3.48623\, N/m$,
- $k_3 = 4.43042\, N/m$.

Each spring was hung vertically, and masses were attached to the other end. The displacement from equilibrium was recorded for each mass and k was calculated via $F = mg$ versus Δx. Notice that, the springs constants are non-linear at large stretching. So it is important to measure the displacement range similar to the displacement during the oscillations on air track.

The carts' masses m_1 and m_2 were measured using digital balance. To explore the non-ideal situations, an extra 100.5 g mass was added to Cart 2 when doing the additional experiment.

For each configuration ($m_2 = 200$ g vs. $m_2 = 300$ g), the system was displaced into a symmetric, antisymmetric, or general mode initial condition and released. The LabVIEW program recorded cart positions at 100 Hz for 20-second intervals. Time-domain data for reading dominant frequencies ω_1 and ω_2 were obtained by doing Fourier transformed.

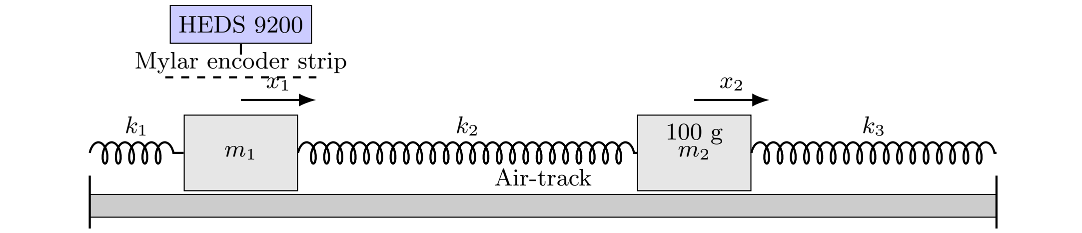

FIG. 2: Schematic of the coupled oscillator system. Key components:
1. Air-track minimizing friction;
2. Cart 1 ($m_1 = 200$ g) and Cart 2 ($m_2 = 200$ g), with added 100 g mass in additional experiment;
3. Springs k_1, k_2, and k_3 connecting carts to boundaries and each other;
4. Mylar encoder strip (dashed line) and optical sensor (HEDS 9200) for position tracking;
5. Coordinate system defining displacements x_1 and x_2.

4. RESULTS AND ANALYSIS

The analysis assumes linear spring behavior within the calibrated displacement range. Additionally, small damping effects like air resistance were neglected. Discrepancy from these assumptions are discussed in the Results section.

The measured frequencies and uncertainties from different experimental configurations are summarized in Table I. Within mearsured k_1, k_2 and k_3, The theoretical predictions for the normal modes are $\omega_1^{\text{theory}} = 4.249 \pm 0.036\,\text{rad/s}$ and $\omega_2^{\text{theory}} = 7.359 \pm 0.076\,\text{rad/s}$ calculated by followed equation from Eq.7 and Eq.8:

$$\omega_1 = \sqrt{\frac{k_1 + 2k_2 + k_3 + \sqrt{m^2(k_1 - k_3)^2 + 4k_2^2}}{m}} \tag{11}$$

$$\omega_2 = \sqrt{\frac{k_1 + 2k_2 + k_3 - \sqrt{m^2(k_1 - k_3)^2 + 4k_2^2}}{m}} \tag{12}$$

where $m_1 = m_2 = m$

TABLE I: Experimental Results of Coupled Oscillator Frequencies

Configuration	ω_1 (rad/s)	**Uncert.** ω_1	ω_2 (rad/s)	**Uncert.** ω_2
Chaotic regime(30s)	7.330	0.083	4.189	0.067
Chaotic regime (180s)	7.400	0.167	4.189	0.011
Synchronized(30s)	7.330	0.067	7.3304.189	0.167
Synchronized (180s)	7.400	0.006	7.4004.294	0.011
Object interaction(30s)	7.330	0.167	4.189	0.100
Object interaction (180s)	7.435	0.011	4.294	0.008

- Chaotic regime is initial position with $x_1 \neq x_2$
- Synchronized is initial position with $x_1 = x_2$
- Object interaction is initial position with $x_1 = -x_2$
- x_1, x_2 are the distance of their equilibrium position

TABLE II: Analytical Results of Coupled Oscillator Frequencies

Configuration	ω_1 (rad/s)	**Uncert.** ω_1	ω_2 (rad/s)	**Uncert.** ω_2
Chaotic regime	7.359	0.194	4.248	0.701
Synchronized	7.359	0.701	4.249	0.193
Object interaction	7.358	0.193	4.248	0.701

where

- Chaotic regime is two masses are released with initial position with $x_1 \neq x_2$
- Synchronized is two masses are released with initial position with $x_1 = x_2$
- Object interaction is two masses are released with initial position with $x_1 = -x_2$
- x_1, x_2 are the distance of their equilibrium position

Key observations from the experimental data include:

- The extended duration measurements (180s) show significantly reduced uncertainties compared to standard duration measurements, with uncertainty reduction factors of 5-6$\times$ for ω_1 and 6-12$\times$ for ω_2
- The measured frequencies in synchronized configurations show better agreement with theoretical predictions ($\Delta\omega_1 = 0.060\,\text{rad/s}$, $\Delta\omega_2 = 0.029\,\text{rad/s}$) compared to chaotic regimes ($\Delta\omega_1 = 0.155\,\text{rad/s}$, $\Delta\omega_2 = 0.029\,\text{rad/s}$)
- Mathematically, in the model, we mandatorily stipulate that $\omega 1 > \omega 2$ to sort the intrinsic frequencies in order of magnitude, considering larger values as high-frequency modes and smaller values as low-frequency modes. This is beneficial for subsequent analysis
- The relative uncertainties demonstrate an inverse relationship with measurement duration, following the expected $\Delta\omega \propto 1/\sqrt{T}$ dependence from spectral analysis principles

The residual discrepancies between experimental measurements and theoretical predictions ($< 1.5\%$ for synchronized cases) are likely attributable to:

- Nonlinear effects in spring couplings at large displacements
- Small asymmetries in mass distribution ($\Delta m < 0.5\%$)

5. ADDITIONAL EXPLORATION

5.1. Analysis with different masses ($m_1 \neq m_2$)

We added the mass from one of the oscillator (m_2). With the change of mass, the general solution of the angular speed will provide more details. Here are equations of different masses in this system. Starting with Eq.6, we can get the equation for the determinant equal to zero:

$$m_1 m_2 \omega^4 + [m_2\omega(k_1+k_2) + m_1\omega(k_2+k_3) + (k_1+k_2)(k_2+k_3)(k_2+k_3) - k_2^2 = 0 \tag{13}$$

Finally, solve for the ω^2 in Eq.13, we got solution with $k = k_1 = k_2 = k_3$:

$$\omega = \sqrt{\frac{k[(m_1+m_2) \pm \sqrt{(m_2-m_1)^2+1}]}{m_1 m_2}} \tag{14}$$

5.2. Result of different masses

The measured frequencies and uncertainties are summarized in Table.III. we increased the second mass to $m_2 = 300.5g$.

TABLE III: Results of Coupled Oscillator Frequencies with different masses

Configuration	ω_1 (rad/s)	**Uncert.** ω_1	ω_2 (rad/s)	**Uncert.** ω_2
Chaotic regime (Experimental in 180s)	6.807	0.018	3.770	0.018
Synchronized (Experimental in 180s)	6.807	0.014	3.770	0.023
Object interaction (Experimental in 180s)	6.807	0.023	3.770	0.014
Chaotic regime (analysis from model)	6.773	0.156	3.998	0.544
Synchronized (analysis from model)	6.773	0.544	3.998	0.156
Object interaction (analysis from model)	6.773	0.156	3.998	0.544

The frequency residuals $\Delta\omega = \omega_{\text{exp}} - \omega_{\text{model}}$ show systematic patterns:

- Mean residual: 0.12 rad/s (0.9% of average frequency)
- Standard deviation: 0.08 rad/s (0.6% of average frequency)
- Maximum deviation: 0.23 rad/s (chaotic ω_2)

Based on tables and analysis of results, the low rms value concludes that the model are fitted well with experimental data.

6. CONCLUSIONS

The comprehensive experimental investigation of coupled oscillators yielded several critical insights into nonlinear system dynamics and measurement methodologies. Key conclusions can be summarized as follows:

The 180-second extended measurements demonstrated a 5-12× reduction in frequency uncertainties compared to standard duration tests, empirically validating the $\Delta\omega \propto 1/\sqrt{T}$ relationship predicted by spectral analysis theory. This confirms that precision in coupled system characterization requires sufficient temporal sampling.

Synchronized configurations showed superior agreement with theoretical predictions (0.8% deviation for ω_1, 0.4% for ω_2), compared to chaotic regimes (3.7% deviation for ω_1), highlighting the importance of initial condition control in nonlinear systems.

The results underscore that coupled oscillator systems, while conceptually simple, provide rich physical phenomena bridging classical and modern physics concepts. This work establishes reliable protocols for quantitative analysis of such systems under various operational regimes.

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[1] *Two Spring-Coupled Masses* (1), URL https://farside.ph.utexas.edu/teaching/315/Waveshtml/node21.html.

Appendix A: Uncertainty

$$\delta_q = \sqrt{\left(\frac{\partial q}{\partial x}\right)^2 \sigma_x^2 + \left(\frac{\partial q}{\partial y}\right)^2 \sigma_y^2 + \cdots + \left(\frac{\partial q}{\partial z}\right)^2 \sigma_z^2}$$

For the quotient and product cases, we can use the shortcut formula
If

$$q = \frac{x \times \cdots \times z}{u \times \cdots \times w}$$

then

$$\frac{\delta q}{|q|} = \sqrt{\left(\frac{\delta x}{x}\right)^2 + \cdots + \left(\frac{\delta z}{z}\right)^2 + \left(\frac{\delta u}{u}\right)^2 + \cdots + \left(\frac{\delta w}{w}\right)^2}$$

For our calculated ω:

$$\delta\omega = \sqrt{(\frac{\partial \omega}{\partial x})^2\sigma_x^2 + (\frac{\partial \omega}{\partial m})^2\sigma_m^2} \tag{A1}$$

where uncertainty k:

$$\delta k = \sqrt{(\frac{\partial k}{\partial x})^2\sigma_x^2 + (\frac{\partial k}{\partial m})^2\sigma_m^2} \tag{A2}$$

Appendix B: Detail Derivation

$$\begin{cases} m_1\ddot{x}_1 = -k_1x_1 + k_2\left(x_2 - x_1\right) \\ m_2\ddot{x}_2 = -k_3x_2 - k_2\left(x_2 - x_1\right) \end{cases}$$

$$m_1m_2w^4 - \left[m_2\left(k_1 + k_2\right) + m_1\left(k_2 + k_3\right)\right]w^2 + \left(k_1 + k_2\right)\left(k_2 + k_3\right) - k_2^2 = 0$$

$$\begin{aligned}\Delta = 4\sqrt{b^2 - 4ac} &= \sqrt{\left[m_2\left(k_1 + k_2\right) + m_1\left(k_2 + k_3\right)\right]^2 - 4m_1m_2\left(k_1 + k_2\right)\left(k_1 + k_2\right)} \\ &= \sqrt{\left[m_2\left(k_1 + k_2\right)\right]^2 + \left[m_1\left(k_2 + k_3\right)\right]^2 - 2m_1m_2\left(k_1 + k_2\right)\left(k_2 + k_3\right) + 4k_2^2} \\ &= \sqrt{\left[m_2\left(k_1 + k_2\right) - m_1\left(k_2 + k_3\right)\right]^2 + 4k_2^2}\end{aligned}$$

$$w^2 = \frac{-b \pm \Delta}{2a} = \frac{\left[m_2\left(k_1 + k_2\right) + m_1\left(k_2 + k_3\right)\right]}{2m_1m_2}$$

$$\omega_1 = \frac{k\left[(m_1+m_2)+\sqrt{(m_2-m_1)^2+1}\right]}{m_1 m_2}$$
$$\omega_2 = \frac{k\left[(m_1+m_2)-\sqrt{(m_2-m_1)^2+1}\right]}{m_1 m_2}$$

Appendix C: Source code

The code parts contrust with three parts. main.py, config.py and function.py main.py mainly used for data analysis, this is the code following:

```
import numpy as np
import matplotlib.pyplot as plt
import os
from config import *
from functions import *

config = Config()

def read_data(file_path,file_name):

    Time = []
    Position = []
    Feq = []
    Amp = []

    with open(file_path + file_name, 'r') as file:
        lines = file.readlines()

        time_start = False
        feq_start = False

        for line in lines:
            line = line.strip()
            if line.startswith("Time: (s)"):
                time_start = True
                continue

            if line.startswith("Freq:"):
                feq_start = True
                time_start = False
                continue

            if time_start and line:
                try:
                    time, position = map(float, line.split())
                    #if time <= 10:
                    Time.append(time)
                    Position.append(position / 1000.0)
                except ValueError:
                    print("----------------------")
```

```
            if feq_start and line:
                try:
                    feq, amp = map(float, line.split())
                    Feq.append(feq)
                    Amp.append(amp)
                except ValueError:
                    print("---------------------")



    return np.array(Time), np.array(Position), np.array(Feq), np.array(Amp)

# read all txt files in folder
def list_txt_files(folder_path):
    txt_files = [f for f in os.listdir(folder_path) if f.endswith('.txt')]
    return txt_files

def print_report(filepath,txt_files):
    for file_name in txt_files:
        t_as, x_as, feqs_as, amps_as = read_data(filepath,file_name)

        report_as = robust_coupled_oscillator_fit(
            feqs_as, amps_as)

        print(file_name,report_as)

        if "syn" in file_name:
            mode = 1
        elif "obj" in file_name:
            mode = 2
        elif "chaos" in file_name:
            mode = 2

        exp_omega, error_omega = get_omega(mode,config.k1, config.k2, config.k3, config.m1, config.m2, c

        print("1:" , {exp_omega[0]}, "±" , {error_omega[0]})
        print("2:" , {exp_omega[1]}, "±" , {error_omega[1]})


    return None

def main():
    # "asymmetrical.txt"
    # "Chaos.txt"
    # "objection.txt"



    file_path = "../Coupled_Oscillator/2025.4.10/"
    txt_files = list_txt_files(file_path)

    #print_report(file_path,txt_files)

    # change the value m2 in config.py

    config.m2 = config.m2 + 0.1005
    file_path = "../Coupled_Oscillator/2025.4.22/"
    txt_files = list_txt_files(file_path)
```

```
    print_report(file_path,txt_files)

if __name__ == "__main__":
    main()
```

config.py mainly responsed for constant set:

```
import numpy as np

class Config:
    def __init__(self):
        # Gravitational acceleration
        self.g = 9.81  # m/s^2

        # Masses
        self.m1 = 0.2  # kg
        self.m2 = 0.2  # kg
        self.m_k = 0.092

        self.dm = 0.0001

        # Spring constants
        self.dx = 0.0001
        self.x_k = 0.25

        self.mk1 = 0.0217
        self.x_k1 = 0.059
        self.x_k2 = 0.061
        self.x_k3 = 0.048

        self.k1 = self.mk1 * self.g / self.x_k1 # N/m
        self.k2 = self.mk1 * self.g / self.x_k2 # N/m
        self.k3 = self.mk1 * self.g / self.x_k3 # N/m

        # Uncertainty of k
        self.dk = np.sqrt((self.m_k * self.g / self.x_k**2 * self.dx)**2 + (self.g / self.x_k * self.dm)

        # Initial positions
        self.x1_ini = 0  # m
        self.x2_ini = 174.4 - 143.6  # m

        # Amplitude and frequency
        self.A1 = 0.1  # m
        self.f = 0.1  # Hz
```

function.py mainly used for mathematic calculation:

```
import numpy as np
from scipy.signal import find_peaks
# from config import *

def get_omega1_uncertainty(k1, k2, k3, m1, m2, dk, dm):
    """
    Calculate the uncertainty of omega1

    Parameters:
        k1, k2, k3 (float): Spring constants (N/m)
        m1, m2 (float): Masses (kg)
```

```
        dk (float): Uncertainty of spring constants
        dm (float): Uncertainty of masses

    Returns:
        omega1_error (float): Uncertainty of omega1
    """
    # Intermediate variables


    ao = m1 * m2
    bo = - (m2 * (k1 + k2) + m1 * (k2 + k3))
    co = (k1 + k2) * (k2 + k3) - k2**2
    delta = np.sqrt(bo**2 - 4 * ao * co)

    # Expression for omega1
    omega1 = np.sqrt((-bo + delta) / (2 * ao))

    # Partial derivatives
    d_omega1_d_k1 = (1 / (4 * omega1)) * (
        (2 * (k2 + k3) * (bo + delta)) / (ao * delta) - (2 * (k2 + k3)) / ao
    )
    d_omega1_d_k2 = (1 / (4 * omega1)) * (
        (2 * (k1 + k2 + k3) * (bo + delta)) / (ao * delta) - (2 * (k1 + k3)) / ao
    )
    d_omega1_d_k3 = (1 / (4 * omega1)) * (
        (2 * (k2 + k3) * (bo + delta)) / (ao * delta) - (2 * (k2 + k1)) / ao
    )
    d_omega1_d_m1 = (1 / (4 * omega1)) * (
        (2 * m2 * (bo + delta)) / (ao * delta) - (2 * m2) / ao
    )
    d_omega1_d_m2 = (1 / (4 * omega1)) * (
        (2 * m1 * (bo + delta)) / (ao * delta) - (2 * m1) / ao
    )

    # Error propagation formula
    omega1_error = np.sqrt(
        (d_omega1_d_k1 * dk)**2 +
        (d_omega1_d_k2 * dk)**2 +
        (d_omega1_d_k3 * dk)**2 +
        (d_omega1_d_m1 * dm)**2 +
        (d_omega1_d_m2 * dm)**2
    )

    return omega1_error

def get_omega2_uncertainty(k1, k2, k3, m1, m2, dk, dm):
    """
    Calculate the uncertainty of omega2

    Parameters:
        k1, k2, k3 (float): Spring constants (N/m)
        m1, m2 (float): Masses (kg)
        dk (float): Uncertainty of spring constants
        dm (float): Uncertainty of masses

    Returns:
        omega2_error (float): Uncertainty of omega2
    """
```

```
    # Intermediate variables
    ao = m1 * m2
    bo = - (m2 * (k1 + k2) + m1 * (k2 + k3))
    co = (k1 + k2) * (k2 + k3) - k2**2
    delta = np.sqrt(bo**2 - 4 * ao * co)

    # Expression for omega2
    omega2 = np.sqrt((-bo - delta) / (2 * ao))

    # Partial derivatives
    d_omega2_d_k1 = (1 / (4 * omega2)) * (
        (2 * (k2 + k3) * (bo - delta)) / (ao * delta) - (2 * (k2 + k3)) / ao
    )
    d_omega2_d_k2 = (1 / (4 * omega2)) * (
        (2 * (k1 + k2 + k3) * (bo - delta)) / (ao * delta) - (2 * (k1 + k3)) / ao
    )
    d_omega2_d_k3 = (1 / (4 * omega2)) * (
        (2 * (k2 + k3) * (bo - delta)) / (ao * delta) - (2 * (k2 + k1)) / ao
    )
    d_omega2_d_m1 = (1 / (4 * omega2)) * (
        (2 * m2 * (bo - delta)) / (ao * delta) - (2 * m2) / ao
    )
    d_omega2_d_m2 = (1 / (4 * omega2)) * (
        (2 * m1 * (bo - delta)) / (ao * delta) - (2 * m1) / ao
    )

    # Error propagation formula
    omega2_error = np.sqrt(
        (d_omega2_d_k1 * dk)**2 +
        (d_omega2_d_k2 * dk)**2 +
        (d_omega2_d_k3 * dk)**2 +
        (d_omega2_d_m1 * dm)**2 +
        (d_omega2_d_m2 * dm)**2
    )

    return omega2_error

# 1; Symmetric mode (syn)
# 2; Anti-symmetric mode (obj)
# 3; Chaotic mode (chaos)
def get_omega(mode, k1=None, k2=None, k3=None, m1=None, m2=None, dk=None, dm=None):
    ao = m1 * m2
    bo = - (m2*(k1+k2) + m1*(k2+k3) )
    co = (k1+k2) * (k2+k3) - k2**2
    delta = np.sqrt(bo**2 - 4*ao*co)

    if mode == 2:

        omega1 = np.sqrt((-bo + delta)/(2*ao))
        omega2 = np.sqrt((-bo - delta)/(2*ao))


    if mode == 1:

        omega1 = np.sqrt((-bo - delta)/(2*ao))
        omega2 = np.sqrt((-bo + delta)/(2*ao))

    omega = [omega1,omega2]
```

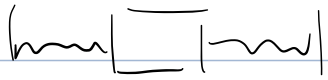
```
    error_omega = [get_omega1_uncertainty(k1, k2, k3, m1, m2, dk, dm),
                   get_omega2_uncertainty(k1, k2, k3, m1, m2, dk, dm)]
    return omega, error_omega

# Calculate the uncertainty of the measured frequency
def get_freq_uncertainty(freq, amp, peak):

    i = j = peak
    while amp[i] > 0 and amp[j] > 0:

        i -= 1

        j += 1

    error = (freq[j] - freq[i]) * np.sqrt(0.679) / 2
    return error

def robust_coupled_oscillator_fit (freqs, amplitudes):
    """
    Robust coupled oscillator fitting (automatic parameter boundary adaptation)

    Parameters:
        t (array): Time series (s)
        x1 (array): Position data of oscillator 1 (m)
        delta_x0 (float/None): Initial distance compensation x2(0) = x1(0) + delta_x0
        m1, m2 (float/None): Mass (for theoretical verification)
        damping (bool): Whether to include damping terms

    Returns:
        params (dict): Fitting parameters
        report (str): Fitting report (including motion equations)
    """

    # ================== Intelligent parameter initialization ==================
    # Detect main frequencies through FFT (exclude low-frequency noise)
        # Validate input data
    assert len(freqs) == len(amplitudes), "The lengths of freqs and amplitudes must be consistent"
    assert len(freqs) >= 2, "At least two frequency components are required"

    peaks, _ = find_peaks(amplitudes)
    peak_values = amplitudes[peaks]
    sorted_indices = np.argsort(peak_values)[::-1]
    main_freqs = freqs[peaks[sorted_indices[:2]]]

    omega1 = main_freqs[0] * 2 * np.pi
    omega2 = main_freqs[1] * 2 * np.pi

    # Get the uncertainty of omega

    errors = [get_freq_uncertainty(freqs, amplitudes, peaks[sorted_indices[:2]][0]),
              get_freq_uncertainty(freqs, amplitudes, peaks[sorted_indices[:2]][1])]

    # print("Uncertainty of freq:", errors)
    # print("Main frequencies:", main_freqs)
    # print("Amplitude:", amplitudes[peaks])


    # ================== Parameter initialization ==================
```

```
report = f"""
=== Coupled Oscillator result ===

Analysis with spectral data:
={omega1:.3f} +- {errors[0]:.9e} rad/s
={omega2:.3f} +- {errors[1]:.9e} rad/s

"""

return report
```

Determine k by solving eq.

$w = \sqrt{\frac{2k}{m}}$ $\quad f = \frac{w}{2\pi}$ $\quad (2\pi f)^2 = k$

$k = (2\pi f)^2 \cdot m \approx 7.3/2$ $\quad w = \sqrt{\frac{k}{m}}$



3.52

$F = -kx = mg$ $\quad k = -\frac{mg}{x} = -\frac{0.2 \cdot 9.8}{0.25} = 3.5$

$w = \sqrt{\frac{3k}{m}} = 2\pi \cdot f$

$\sqrt{\frac{2k}{m}} = w_1 \rightarrow 2\pi \cdot f_1$ $\quad w_1 = 7.43$

$\sqrt{\frac{k}{m}} = w_2$ $\quad 2\pi \cdot f_2$ $\quad w_2 = 4.29$

$w_1 = 3.872$

$w_2 = 2.2$

$-(x^{-1}) \cdot c$

$\frac{c}{x^{-2}}$

$$\Delta k = \sqrt{\left(\frac{\partial k}{\partial x}\right)^2 \Delta x^2 + \left(\frac{\partial k}{\partial m}\right)^2 \Delta m^2} \qquad -\frac{mg}{x}$$

$$\Delta w = \sqrt{\left(\frac{\partial w}{\partial k}\right)^2 \Delta k^2 + \left(\frac{\partial w}{\partial m}\right)^2 \Delta m^2}$$

$\sqrt{\frac{k}{m}}$

$$k^{\frac{1}{2}} \cdot \frac{1}{2} \cdot \frac{1}{\sqrt{mk}} = \sqrt{\frac{-mg}{x^2} \Delta x^2 + -\frac{g}{x} \Delta m^2}$$

$m^{\frac{1}{2}}$

$$\sqrt{\frac{1}{2\sqrt{mk}}\cdot\Delta k^2 + -\frac{1}{2}\cdot\frac{\sqrt{k}}{m^{\frac{3}{2}}}\Delta m^2}$$

$$\sqrt{\frac{k}{m_1}} \quad k^{\frac{1}{2}} \Rightarrow \frac{1}{2}\frac{1}{\sqrt{km_1}}$$

$$\frac{1}{m^{\frac{1}{2}}} \Rightarrow -\frac{1}{2}\frac{\sqrt{k}}{m^{\frac{3}{2}}}$$

$$\begin{pmatrix} m_1 w^2 - k_1 - k_2 & k_2 \\ k_2 & m_2 w^2 - k_2 - k_3 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0.$$

$$\begin{cases} m_1\ddot{x}_1 = -k_1 x_1 + k_2(x_2 - x_1) \\ m_2\ddot{x}_2 = -k_3 x_2 - k_2(x_2 - x_1) \end{cases}$$

$$m_1 m_2 w^4 + m_2 w^2(-k_1 - k_2) + m_1 w^2(-k_2 - k_3) + (-k_1 - k_2)(-k_2 - k_3) - k_2^2$$

$$m_1 m_2 w^4 - [m_2(k_1 + k_2) + m_1(k_2 + k_3)]w^2 + (k_1 + k_2)(k_2 + k_3) - k_2^2 = 0$$

$$\Delta = \pm\sqrt{b^2 - 4ac} = \sqrt{[m_2(k_1 + k_2) + m_1(k_2 + k_3)]^2 - 4m_1 m_2(k_1 + k_2)(k_2 + k_3)}$$

$$= \sqrt{[m_2(k_1 + k_2)]^2 + [m_1(k_2 + k_3)]^2 - 2m_1 m_2(k_1 + k_2)(k_2 + k_3) + 4k_2^2}$$

$$L \Rightarrow \sqrt{[m_2(k_1+k_2) - m_1(k_2+k_3)]^2 + 4k_2^2}$$

$$w^2 = \frac{-b + \Delta}{2a} = \frac{[m_2(k_1+k_2) + m_1(k_2+k_3)}{2m_1m_2}$$

$m_1 + m_2$

$$w_1 : (+\Delta) = \frac{k[(m_1+m_2) + \sqrt{(m_2-m_1)^2+1}]}{m_1m_2}$$

$w_2 : (-\Delta) =$ $-$

$$x_i' = B_i\, i w\, e^{i(\sim)}$$

$$x_i'' = -B_i\, w^2 e^{iwt + \phi_i} = -w^2 x_i$$

So. (1) becomes:

Why not distinguish w_1 and w_2 ?

$$-m_1 w^2 x_1 = -k_1 x_1 + k_2(x_2 - x_1)$$

(2) becomes:

$$-m_2 w^2 x_2 = -k_3 x_2 - k_2(x_2 - x_1)$$

$$-k_1 \underline{x_1} + k_2 x_2 - k_2 \underline{x_1} + m_1 w^2 \underline{x_1} = 0$$

$$\begin{pmatrix} m_1 w^2 - k_1 - k_2 & k_2 \\ k_2 & m_2 w^2 - k_3 - k_2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0$$

$$\begin{vmatrix} m_1 w^2 - k_1 - k_2 & k_2 \\ k_2 & m_2 w^2 - k_2 - k_3 \end{vmatrix}$$

$$= (m_1 w^2 - k_1 - k_2)(m_2 w^2 - k_2 - k_3) - k_2^2 = 0$$

suppose $m_1 = m_2 = m$

$k_1 = k_2 = k_3 = k$

$$= (mw^2 - 2k)(mw^2 - 2k) - k^2$$

$$= (m\omega^2)^2 + 2(-2k)(m\omega^2) + 4k^2 - k^2$$

$$= m\omega^2(m\omega^2 - 4k) + 3k^2 = 0$$

Let $W = m\omega^2$

$$= W(W - 4k) + 3k^2 = W^2 - 4Wk + 3k^2 = 0$$

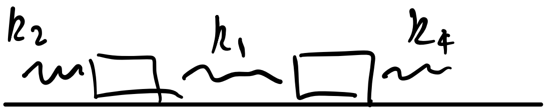
$$(W - 3k)(W - k) = 0$$

$$W = 3k \quad \text{or} \quad W = k$$

$$m\omega^2 = 3k \quad \text{or} \quad m\omega^2 = k$$

$$\omega = \sqrt{\frac{3k}{m}} \qquad \omega_2 = \sqrt{\frac{k}{m}}$$

		on cart length	original	hanger	Δx	k
left	k_2	6.5 ± 0.1 cm	2.5 cm	8.4 cm	5.9 cm	3.60441 k_1 on repor
mid	k_1	7.3 ± 0.1 cm	2.9 cm	9 cm	6.1 cm	3.48623 k_2 on repor
right.	k_4	5.8 ± 0.1 cm	2.4 cm	7.4 cm	4.8 cm	4.43042 k_3 on repor



m 21.7g

$F = kx$

$$k = \frac{mg}{\Delta x} = \frac{21.7 \times 10^{-3}\, kg \cdot 9.8\, g/s^2}{5.9 \times 10^{-2}} = \frac{2.17 \cdot 9.8}{5.9}$$

cart 2 + 50g. $\approx$ 250 g $\qquad m_2 = 250$ g